

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

192511445

I SHAIK KHATIJAH ROSHAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shai

20/20

Shai

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign

suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Co2-AT-1

ANALYTICAL PROBLEMSOLVING

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1. Given:

Network Address = 192.168.20.0/24

Lab A = 126

Lab B = 127

Lab C = 128

Step: 1 - Find the number of hosts,

$$\text{usable Hosts} = 2^{(32 - \text{prefix})} - 2$$

Lab A (126):

$$\text{Host bits} = 32 - 26 = 6$$

$$\text{Total addresses} = 2^6 = 64$$

$$\text{usable hosts} = 64 - 2 = 62$$

Block size = 64

subnet starts at 0

Item	Value
Network Address	192.168.20.0
First Host	192.168.20.1
Last Host	192.168.20.62
Broadcast	192.168.20.63
usable Hosts	62

Lab B (127):

Host bits = 5

Total addresses = 32

usable = 30

Next available subnet starts after 63

Network = 192.168.20.64

Item	Value
Network Address	192.168.20.64
First Host	192.168.20.65
Last Host	192.168.20.94
Broadcast	192.168.20.95
usable Hosts	30

Lab C (128):

Host bits = 4

Total addresses = 16

usable = 14

Next subnet begins at 96

Item	Value
Network Address	192.168.20.96
First Host	192.168.20.97
Last Host	192.168.20.110
Broadcast	192.168.20.111
usable Hosts	14

Final Answer :

Lab	Network	Host Range	Broad cast	usable Hosts
A	192.168.20.0/26	1-62	63	62
B	192.168.20.64/27	65-94	95	30
C	192.168.20.96/28	97-110	111	14

2, Given :

System : 1

IP : 192 168 1 130

Mask : 255 255 255 192 (126)

System : 2

IP : 192.168.1.190

Mask : 255 255 255 192 (126)

Step - 1: Find Block size

$$256 - 192 = 64$$

Subnets are

$$\Rightarrow 0 - 63$$

$$\Rightarrow 64 - 127$$

$$\Rightarrow 128 - 191$$

$$\Rightarrow 192 - 255$$

Both IPs lie between 128-191.

System : 1

Network Address = 192.168.1.128

Broad cast = 192.168.1.191.

Host Range

192.168.1.129 - 192.168.1.190

usable Hosts

$$2^6 - 2 = 62$$

Since 130 lies within the host range,
It belongs to this subnet.

System: 2

Network Address = 192.168.1.128

Broadcast = 192.168.1.191

Host Range

192.168.1.129 - 192.168.1.190

usable Hosts = 62

190 lies in the host range.

Hence, it belongs to the subnet.

Final Answer:

Item	system 1	system 2
Network	192.168.1.128	192.168.1.128
Host Range	129.190	129.190
Broadcast	191	191
usable Hosts	62	62
Valid IP?	yes	yes

Result: Both systems belong to the same subnet.

3, Given :

Network = 172.16.0.0/16

Departments

HR → 125

Sales → 126

IT → 127

Admin → 128

HR (125)

Addresses = 128

usable = 126

Item	Value
Network	172.16.0.0
Host Range	172.16.0.1 - 172.16.0.126
Broad cast	172.16.0.127
Hosts	126

Sales (126)

starts after HR

Item	Value
Network	172.16.0.128
Host Range	172.16.0.129 - 172.16.0.190
Broad cast	172.16.0.191
Hosts	62

IT (127)

Item	Value
Network	172.16.0.192
Host Range	172.16.0.193-172.16.0.222
Broadcast	172.16.0.223
Hosts	30

Admin (128)

Item	Value
Network	172.16.0.224
Host Range	172.16.0.225-172.16.0.238
Broadcast	172.16.0.239
Hosts	14

Remaining unused Range.

starts after 172.16.0.239

Remaining:

172.16.0.240 - 172.16.255.255

4, Given:

Network = 172.16.10.0/24

Mask = /27

Step: 1

Borrowed Bits

$$27 - 24 = 3$$

$$\text{Subnets} = 2^3 = 8$$

Step : 2

Hosts Bits

$$32 - 27 = 5$$

$$\text{Hosts} = 2^5 - 2$$
$$= 30$$

Step : 3

Block size

$$256 - 224 = 32$$

Subnets

Network	Broadcast
0	31
32	63
64	95
96	127
128	159
160	191
192	223
224	255

Step : 4

IP : 172.16.10.78

78 lies between

64-95

Therefore

Network = 172.16.10.64

Broadcast = 172.16.10.95

Host Range

65-94

Step : 5

IP : 172.16.10.95

95 is the broadcast address

Hence,

It belongs to the same subnet cannot be assigned to a host.

Final Answer:

⇒ Total subnets = 8

⇒ usable hosts = 30

⇒ Network Address = 172.16.10.64

⇒ Broadcast = 172.16.10.95

⇒ 172.16.10.78 is a valid host in this subnet.

⇒ 172.16.10.95 is the broadcast address.

5, Given IPv6 Address

2001:db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the Address

Rules:

Remove leading zeros.

Replace consecutive zero groups with "::" (once only)

compressed form:

2001:db8::ff00:42:8329

Step 2: Expand the Address

Expanded form

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Find network prefix (/64)

First 64 bits (first four groups) represent the network.

network prefix 2001:db8:0:0::/64

Step 5: calculate Host Addresses

$$\text{Host bits} = 128 - 64 = 64$$

Total host addresses:

$$= 2^{64}$$

$$= 18,446,744,073,709,551,616$$

[Signature]